

BREAST CANCER DETECTION USING MACHINE LEARNING



Mini Project submitted in partial fulfillment of the requirement for the award of the
degree

BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE AND ENGINEERING

Under the esteemed guidance of

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By

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Cheeryal (V), Keesara (M), Medchal.Dist.-501 301.

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I, **D Sai Bhavani** bearing Roll No. **21R11A05C1** hereby declare that the project report entitled “**Breast Cancer Detection using Machine Learning**” is done under the guidance of **Dr.R.V.Sudhakar, Associate Professor**, Department of Computer Science and Engineering, Geethanjali College of Engineering and Technology, is submitted in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering**.

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ABSTRACT

Breast cancer poses a significant global health challenge, emphasizing the need for accurate tumor classification into malignant or benign categories for effective diagnosis and treatment. This study develops a predictive model for breast cancer classification using logistic regression, a widely-used statistical method. The Breast Cancer Wisconsin (Diagnostic) Dataset, comprising [number] instances and [number] features, is utilized for model development. Logistic regression is employed as the primary algorithm due to its interpretability and efficacy in binary classification tasks. Hyperparameter tuning is performed to optimize model performance and prevent overfitting. Evaluation metrics, including accuracy, precision, recall, and other key performance indicators, are scrutinized to assess the model's effectiveness. A Flask API provides a user-friendly interface for making predictions, enhancing accessibility for healthcare professionals and stakeholders. This study demonstrates the effectiveness of logistic regression in advancing breast cancer diagnosis and treatment strategies, ultimately improving patient outcomes and healthcare decision-making.

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LIST OF ABBREVIATIONS

S. No	Abbreviation	Full Form
1	EDA	Exploratory Data Analysis
2	HTML	Hypertext Markup Language
3	CSS	Cascading Style sheets
4	JS	Java Script
5	API	Application Programming Interface
6	UML	Unified Modeling Language

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1. INTRODUCTION

1.1 ABOUT THE PROJECT

Breast cancer is a major health challenge for women worldwide, with early detection crucial for successful treatment and survival. Traditional methods like mammography and biopsy, while effective, can be invasive and vary in accuracy.

This project leverages machine learning (ML) to enhance breast cancer detection by developing a predictive model using historical medical data. The model will differentiate between malignant and benign tumors by analyzing features such as tumor size and cell shape. Key algorithms like Logistic Regression, Support Vector Machine (SVM), and Random Forest will be used, and model performance will be evaluated with accuracy, precision, recall, and F1-score metrics.

The goal is to improve diagnostic accuracy, contribute to early detection, and ultimately enhance patient outcomes and healthcare efficiency.



Fig 1.1 Image depicting Breast Cancer Awareness

1.2 OBJECTIVES

1. To create a machine learning model capable of accurately predicting the presence of breast cancer based on patient data, including features such as tumor size, cell shape, and texture.
2. To implement and compare the performance of various machine learning algorithms, including Logistic Regression, Support Vector Machine (SVM), and Random Forest, in order to identify the most effective model for breast cancer detection.
3. To improve the accuracy, precision, and recall of breast cancer diagnosis by leveraging machine learning techniques, thereby reducing false positives and false negatives.
4. To preprocess and clean the dataset effectively, ensuring that the input data is of high quality, free from errors, and suitable for training machine learning models.
5. To fine-tune the machine learning models through hyperparameter optimization and cross-validation, achieving the best possible performance for accurate predictions.
6. To develop a user-friendly tool or interface that healthcare professionals can use to input patient data and receive a predictive diagnosis, assisting them in making informed decisions.

2. SYSTEM ANALYSIS

2.1 EXISTING SYSTEM

The current approach to breast cancer detection primarily relies on traditional diagnostic methods, including mammography, ultrasound, magnetic resonance imaging (MRI), and biopsy. These methods are well-established and are considered the standard in clinical practice for identifying and diagnosing breast cancer. Here's a brief overview of the existing techniques:

1. Mammography

Mammography is a specialized medical imaging technique that uses low-dose X-rays to visualize the internal structure of the breast. It is commonly used for routine screening and can detect tumors that are too small to be felt. Mammography is non-invasive, relatively quick, and can detect early signs of breast cancer.

2. Ultrasound

Ultrasound uses high-frequency sound waves to create images of the breast tissue. It is often used as a supplementary tool to mammography, especially in women with dense breast tissue. Ultrasound is also non-invasive, and it helps distinguish between solid tumors and fluid-filled cysts.

3. Magnetic Resonance Imaging (MRI)

MRI uses strong magnetic fields and radio waves to produce detailed images of the breast. It is usually recommended for high-risk patients or for further investigation when mammography results are inconclusive. MRI provides high-resolution images and is particularly useful for evaluating the extent of cancer.

4. Biopsy

Biopsy involves the removal of a small sample of breast tissue for laboratory analysis to determine whether it is cancerous. Biopsy is the definitive method for diagnosing breast cancer, providing conclusive evidence about the nature of the tumor.

DRAWBACKS OF PROPOSED SYSTEM

1. While these methods are essential in the diagnosis and treatment of breast cancer, they have several limitations:
2. While necessary, biopsies are invasive, causing discomfort, anxiety, and potential complications for patients. They can also be time-consuming and require recovery time.
3. Mammography can be uncomfortable for many women, involving compression of the breast, which can cause pain or discomfort. Mammograms can sometimes detect abnormalities that are not cancerous, leading to unnecessary biopsies, anxiety, and further testing. Mammography and ultrasound can miss certain types of breast cancer, particularly in women with dense breast tissue, where the difference between cancerous and normal tissue is less distinct.
4. Although the radiation dose in mammography is low, repeated exposure over time may pose a small risk, especially for women undergoing frequent screenings.
5. MRI is costly, time-consuming, and often inaccessible in low-resource areas, making it typically reserved for high-risk cases. It is very expensive
6. The traditional process, from initial imaging to biopsy and final diagnosis, can be lengthy, delaying the start of treatment and potentially affecting patient outcomes.

2.2 PROPOSED SYSTEM

The proposed system aims to use machine learning (ML) to create an advanced, non-invasive tool for early breast cancer detection. By analyzing patient data to distinguish between malignant and benign tumors, it promises higher accuracy and efficiency compared to traditional methods.

Key Components:

1. Data Collection and Preprocessing

- i. Utilizes datasets like the Wisconsin Breast Cancer Dataset with features from digitized images of breast masses.
- ii. Handles missing values, outliers, and noise.
- iii. Chooses relevant features such as cell size and shape to enhance model performance.

2. Model Development

- i. Includes Logistic Regression, Support Vector Machine (SVM), and Random Forest.
- ii. Splits data into training and testing sets for model evaluation.

3. Performance Evaluation

- i. Accuracy, Precision, Recall, and F1-Score.
- ii. Ensures the model generalizes well to new data.

4. Model Optimization

- i. Uses techniques like Grid Search to find optimal settings.
- ii. Enhances model capabilities with additional features.

5. Deployment and User Interface

- i. Provides an intuitive web or desktop application for healthcare professionals.
- ii. Delivers quick diagnostic results.

6. Integration with Clinical Systems

- i. Integrates with existing Electronic Health Record (EHR) systems.
- ii. Adapts with new data to improve over time.

ADVANTAGES OF PROPOSED SYSTEM

- i. Reduces need for invasive procedures and healthcare costs.
- ii. Enhances diagnostic accuracy and enables earlier detection.
- iii. Extends availability to low-resource settings.
- iv. Provides rapid results, accelerating diagnosis and treatment.
- v. Adapts to include more data sources for improved accuracy

2.3 FEASIBILITY STUDY

2.3.1 DETAILS

- i. The project requires access to relevant datasets, computational resources for model training, and expertise in machine learning and healthcare. The technology and tools needed for development and deployment are available.
- ii. The system is designed to integrate smoothly into existing diagnostic workflows, offering ease of use and improved accuracy to benefit healthcare providers and patients.
- iii. Development and maintenance costs are evaluated, with potential savings from reduced reliance on traditional diagnostics. The project is assessed to fit within the budget and timeline.

2.3.2 Impact on Environment

The system reduces the need for physical procedures and paper records, thereby lowering environmental footprints. Energy consumption and electronic waste from computational resources are considered.

2.3.3 Safety

The system is designed with a strong emphasis on security and patient safety. To protect sensitive patient data, encryption methods are used to ensure that all data transmitted and stored is secure. Access to the system is strictly controlled through secure access mechanisms, meaning only authorized personnel can view or interact with patient data. The system also adheres to industry regulations like HIPAA (Health Insurance Portability and Accountability Act), ensuring that it meets high standards for protecting health information. Regular security audits are conducted to identify and mitigate potential vulnerabilities. In addition, the machine learning models used for diagnosis are rigorously validated to minimize the risk of misdiagnosis, ensuring that the system remains reliable and safe for clinical use.

2.3.4 Ethics

Handling patient data responsibly is at the core of the system's ethical framework. All patient data is collected and processed with informed consent, ensuring transparency and compliance with ethical standards. Biases in the model are continually monitored and minimized to ensure that predictions do not disproportionately affect any specific group or demographic. Ethical standards are maintained throughout the system's development and deployment, ensuring that patient well-being is prioritized and that the system operates within the bounds of medical and societal ethical norms. These practices ensure the system's fairness and prevent any unjust outcomes due to biased data or models.

2.3.5 Cost

The system incurs various costs, including data acquisition, model training, and software development. These upfront costs are essential for developing a robust and accurate diagnostic tool. Additionally, ongoing expenses cover system maintenance, software updates, and future enhancements to ensure the system remains up-to-date with medical and technological advances. Despite the initial investment, the system is designed to significantly reduce costs associated with traditional diagnostic procedures. By improving the accuracy of early breast cancer detection, the system minimizes the number of false positives and negatives, reducing unnecessary treatments and follow-up tests. Over time, this leads to cost savings for both healthcare providers and patients, making the system a cost-effective solution in the long term. The feasibility study confirms that the benefits, such as improved diagnostic accuracy and better patient outcomes, outweigh the costs.

2.3.6 Type

This system functions as a healthcare diagnostic tool specifically aimed at early detection of breast cancer. It leverages machine learning algorithms to analyze patient data and provide predictive insights that help healthcare professionals make informed decisions. The system is scalable and adaptable, meaning it can be implemented in various healthcare settings, from small clinics to large hospitals. Its flexibility allows for integration into different diagnostic workflows, ensuring that it can be tailored to meet the needs of different healthcare institutions while improving the speed and accuracy of breast cancer diagnosis.

2.4 SCOPE OF THE PROJECT

The scope of the **Breast Cancer Detection Using Machine Learning** project includes developing a predictive model to enhance early detection of breast cancer. The project encompasses data collection and preprocessing, machine learning model development, performance evaluation, and optimization. It also involves creating a user-friendly application for healthcare professionals and ensuring integration with existing systems. The focus is on improving diagnostic accuracy and efficiency, with the system designed to be scalable and adaptable to various healthcare environments.

2.5 SYSTEM CONFIGURATION

The "Breast Cancer Detection Using Machine Learning" system requires a robust computational environment to handle data processing, model training, and real-time predictions. The system configuration consists of both hardware and software requirements, ensuring optimal performance and scalability.

Software Required:

1. Operating System-Windows/Mac/Linux
2. Jupyter Notebook
3. Spyder IDE

Hardware Required:

1. GPU integrated Laptop/Desktop
2. RAM: 4GB
3. Hard Disk: 50 GB

3. LITERATURE SURVEY

3.1 LITERATURE REVIEW

The literature review focuses on various machine learning techniques and their application in medical diagnosis, particularly for breast cancer detection. Over the years, numerous studies have explored the use of algorithms like Logistic Regression, Support Vector Machines (SVM), Decision Trees, and Neural Networks to improve accuracy in early cancer detection.

1. **Nasser, Maged, and Umi Kalsom Yusof. "Deep Learning Based Methods for Breast Cancer Diagnosis: A Systematic Review and Future Direction." *Diagnostics* 13, no. 1 (2023): 161.**

This paper presents a systematic review of deep learning methods applied to breast cancer detection, focusing on genomics and histopathological imaging data. It covers various deep learning models, with a particular emphasis on convolutional neural networks (CNNs), which are identified as the most effective models in this domain. The review also discusses common datasets, evaluation metrics, challenges, and future research directions, aiming to provide a comprehensive understanding of the current landscape and trends in deep learning-based breast cancer diagnosis (MDPI). The authors compared six different classifiers to evaluate their performance on a dataset of 3002 mammogram images.

2. **Amin, Farhan, Hussain AlSalman, and Gyu Sang Choi. "Breast Cancer Detection and Prevention Using Machine Learning." *Journal of Imaging* 6, no. 8 (2020): 113.**

This study leverages machine learning models, including CNNs, to enhance breast cancer detection and prevention. The proposed model was tested on a dataset of 3002 mammogram images, achieving high accuracy with reduced computational power. The authors compared six different classifiers and highlighted CNNs' superior performance, emphasizing their potential in early detection and treatment of breast cancer (MDPI).

3. **Divya, A., Aruna B. S., and Manjunath Aradhya V. "Hybrid Deep Learning Model for Breast Cancer Detection." *International Journal of Intelligent Systems and Applications in Engineering* 8, no. 2 (2020): 75-82.**

This paper presents a hybrid deep learning approach combining CNN and support vector machine (SVM) models for breast cancer classification. The model uses image data from mammograms and clinical data to improve diagnostic accuracy. By integrating CNNs for feature extraction, the hybrid model demonstrates enhanced performance compared to traditional methods.

4. **Raza, Asaf, Naeem Ullah, Javed Ali Khan, Muhammad Assam, Antonella Guzzo, and Hanan Aljuaid. "DeepBreastCancerNet: A Novel Deep Learning Model for Breast Cancer Detection Using Ultrasound Images."**

This study proposes a new deep learning model called DeepBreastCancerNet for the detection and classification of breast cancer using ultrasound images. The model consists of 24 layers, including convolutional layers, inception modules, and a fully connected layer, and it achieved an accuracy of 99.35%. The model was validated using a publicly available dataset, achieving the highest accuracy of 99.63%.

5. **Rouzi, Mohammad Dehghan, Behzad Moshiri, Mohammad Khoshnevisan, Mohammad Ali Akhaee, Farhang Jaryani, Samaneh Salehi Nasab, and Myeounggon Lee. "Breast Cancer Detection with an Ensemble of Deep Learning Networks Using a Consensus-Adaptive Weighting Method."**

The study proposes a novel computer-aided detection (CAD) system that leverages an ensemble of deep learning networks including EfficientNet, Xception, MobileNetV2, InceptionV3, and Resnet50. The integration of these networks is managed through a consensus-adaptive weighting (CAW) method, which dynamically adjusts the influence of each network based on their performance. The proposed system was evaluated on multiple datasets, including the DDSM and INbreast, showing a significant improvement in detection rates.

4. SYSTEM DESIGN

4.1 SYSTEM ARCHITECTURE

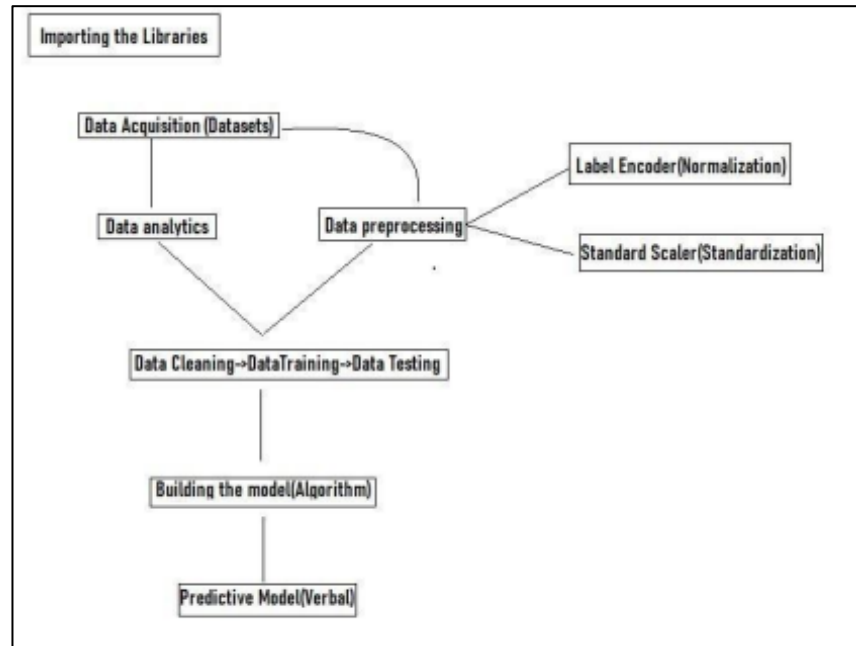


Fig 4.1.1 System Architecture

1. Data Collection

The dataset for this breast cancer diagnosis project was sourced from Kaggle and contains detailed medical records related to breast cancer tumors. Each record is labeled, indicating whether the tumor is malignant (cancerous) or benign (non-cancerous). This labeled nature of the dataset allows for supervised machine learning, where the models can be trained to predict the diagnosis based on the input features. The dataset includes multiple features, such as cell radius, texture, perimeter, and area, which are crucial for determining the nature of the tumor. These diagnostic measures provide the foundation for developing predictive models.

2. Data Pre-processing

To prepare the data for analysis and modeling, several pre-processing steps were undertaken. Initially, the data was structured in CSV format, which made it easily accessible for tools like Pandas, simplifying data handling and manipulation. Data cleaning was a critical step, as any missing or irrelevant information could impact model accuracy. In cases where values were missing, strategies like imputing the mean or removing unnecessary rows or columns were applied, ensuring the dataset was complete and consistent.

3. Data Visualization

Data visualization played a key role in understanding the underlying patterns and relationships between features in the dataset. Tools like Matplotlib and Seaborn were used to generate informative charts and graphs. Visualizing the data helped uncover important insights, such as how features like cell radius or texture correlated with the diagnosis labels. The distributions of these features were visualized through histograms, box plots, and scatter plots, enabling a deeper understanding of the data before applying machine learning techniques. These visualizations allowed for the identification of key features that would be most influential in predicting the diagnosis.

4. Feature Extraction

Feature extraction involved selecting and engineering the most relevant features from the dataset based on their relationship with the target variable, i.e., the diagnosis. By examining the visualized data and calculating statistical correlations, important features like cell radius, perimeter, and texture were identified as strong predictors of whether a tumor was malignant or benign. These features were then used to train machine learning models, such as Logistic Regression, Decision Tree, and Random Forest. By focusing on the most significant attributes, the model's predictive power was improved while minimizing the risk of overfitting or introducing unnecessary complexity.

5. Model Evaluation

Once the machine learning models were trained using the extracted features, they were evaluated for performance and reliability. Cross-validation techniques were applied to prevent overfitting, ensuring that the models would generalize well to unseen data. In this case, k-fold cross-validation was utilized, where the data was divided into k subsets, while the remaining subset was used for validation. Model accuracy was a key metric used to assess performance, calculated by dividing the number of correct predictions by the total number of predictions made.

Algorithms Used

- i. **Logistic Regression:** A binary classification algorithm that predicts probabilities between 0 and 1 for categorical outcomes.
- ii. **Decision Tree:** A model that makes decisions based on a series of branching decisions, used for both classification and regression tasks.
- iii. **Random Forest:** An ensemble method that combines multiple decision trees to improve classification accuracy and reduce overfitting.
- iv. **Support Vector Machine (SVM):** A classification algorithm that finds the hyperplane that best separates classes in the feature space, effective in high-dimensional spaces.
- v. **K-Nearest Neighbors (KNN):** A simple algorithm that classifies data points based on the majority label of their nearest neighbors.
- vi. **XGBoost:** An advanced boosting algorithm that builds models in a sequential manner, improving performance through gradient boosting and regularization.

Steps for Implementation

1. Collect the dataset.
2. Import libraries.
3. Train the model using Logistic Regression, Decision Tree, and Random Forest.
4. Test the model to find the accuracy.
5. Use hyperparameter tuning to optimize the model's performance.

4.2 UML DIAGRAMS

1. Use Case Diagram

The use case diagram for **Breast Cancer Detection** illustrates the interactions between the system and its users. In this case, the primary actors include **Doctors** and **Data Scientists**. For **Doctors**, the use cases include logging in, uploading patient data (like tumor measurements), and reviewing model predictions (malignant or benign). For **Data Scientists**, the use cases include logging in, accessing patient data, training the machine learning model, and evaluating its performance. This diagram provides a high-level view of the system's core functionalities, detailing how different users interact with the system.

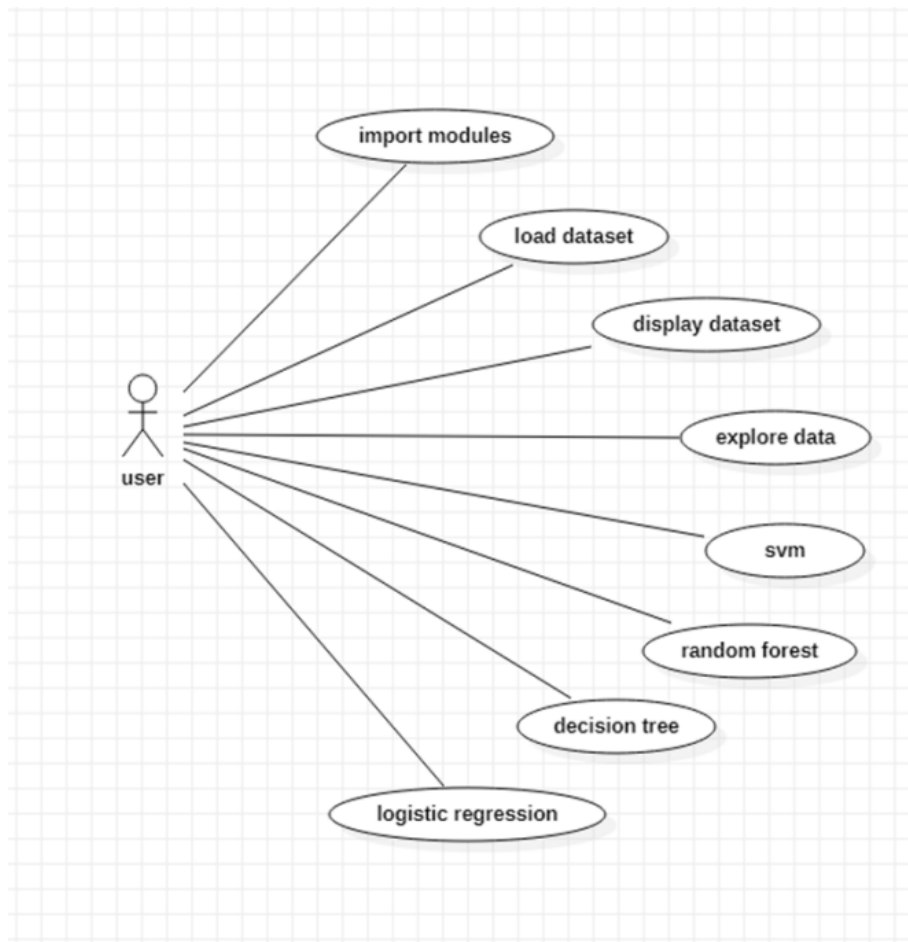


Fig 4.2.1 Use case Diagram

2. Sequence Diagram

The sequence diagram for the **Breast Cancer Detection** system outlines the sequence of interactions between different components of the system during the prediction process. For instance, when a **doctor** uploads patient data, the sequence starts with the doctor initiating a request to upload the tumor attributes. The system processes the request by storing the data and passing it to the machine learning model. The model then classifies the tumor as malignant or benign, and the result is displayed to the doctor. This diagram highlights the flow of interactions between actors over time and how each request moves through the system, illustrating the response of each component.

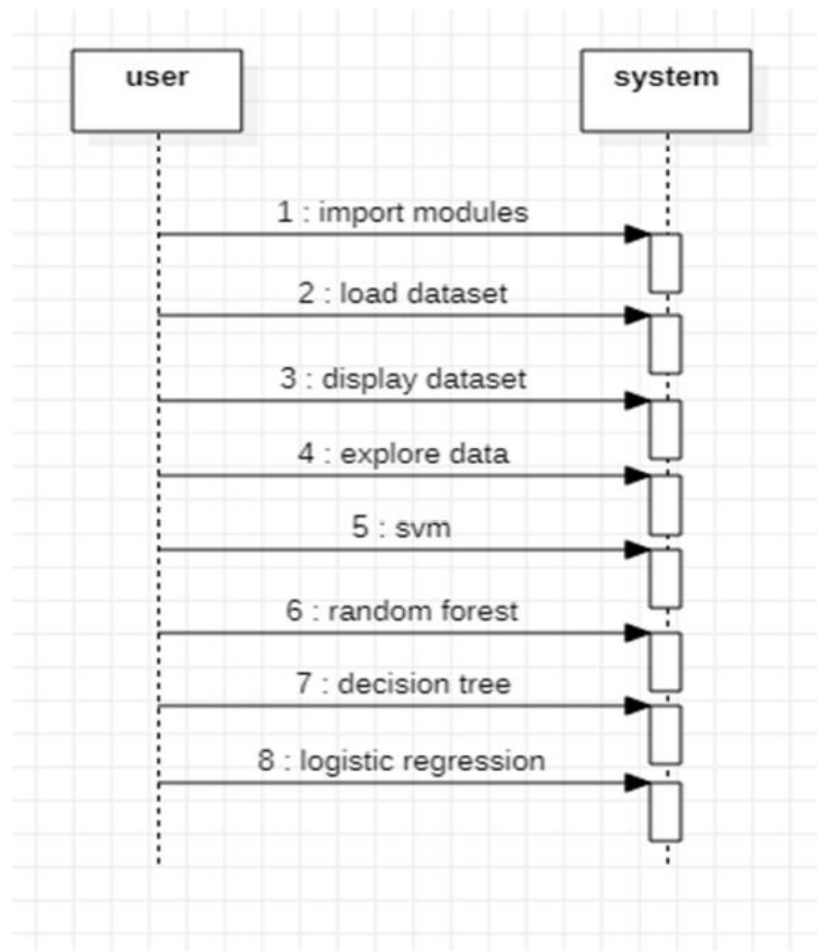


Fig 4.2.2 Sequence Diagram

3. Class Diagram

The class diagram for the **Breast Cancer Detection System** outlines the system's key components, including **Patient**, **Doctor**, and **PredictionModel** classes. The **Patient** class stores patient details and tumor attributes, while the **Doctor** class manages doctor information and facilitates result review. The **PredictionModel** class handles the machine learning model used for classifying tumors as malignant or benign. This diagram provides a clear view of the relationships and interactions between these classes.

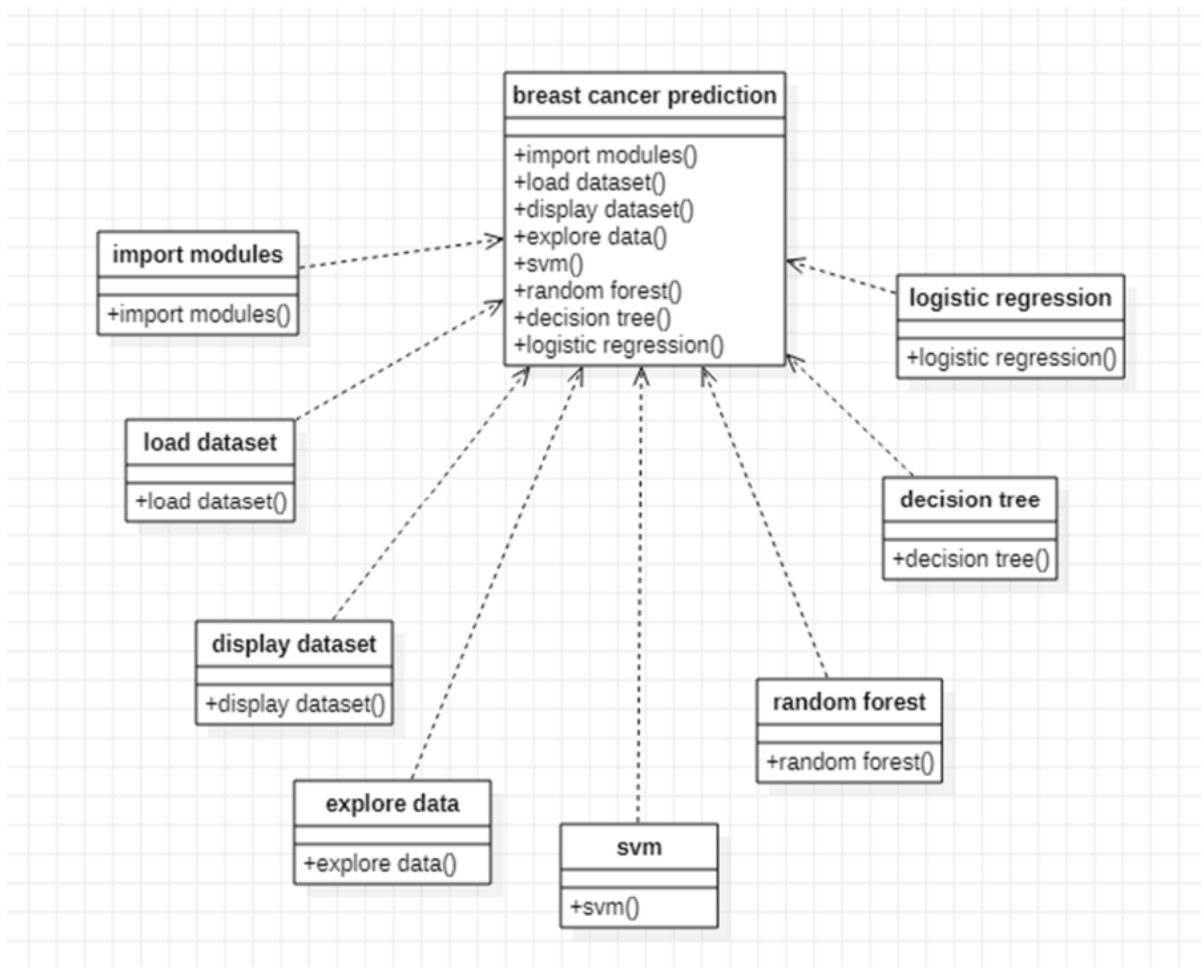


Fig 4.2.3 Class Diagram

4. Activity Diagram

The activity diagram for the **Breast Cancer Detection** system illustrates the workflow for detecting breast cancer from patient data. It begins with a **doctor** entering the tumor attributes into the system. The system then processes this data and passes it to the machine learning model. Based on the input data, the model predicts whether the tumor is malignant or benign. The results are displayed to the doctor, who can review them and take appropriate action. The diagram includes decision points (such as whether the tumor is malignant or benign) and parallel activities, providing a visual representation of the flow of control within the system.

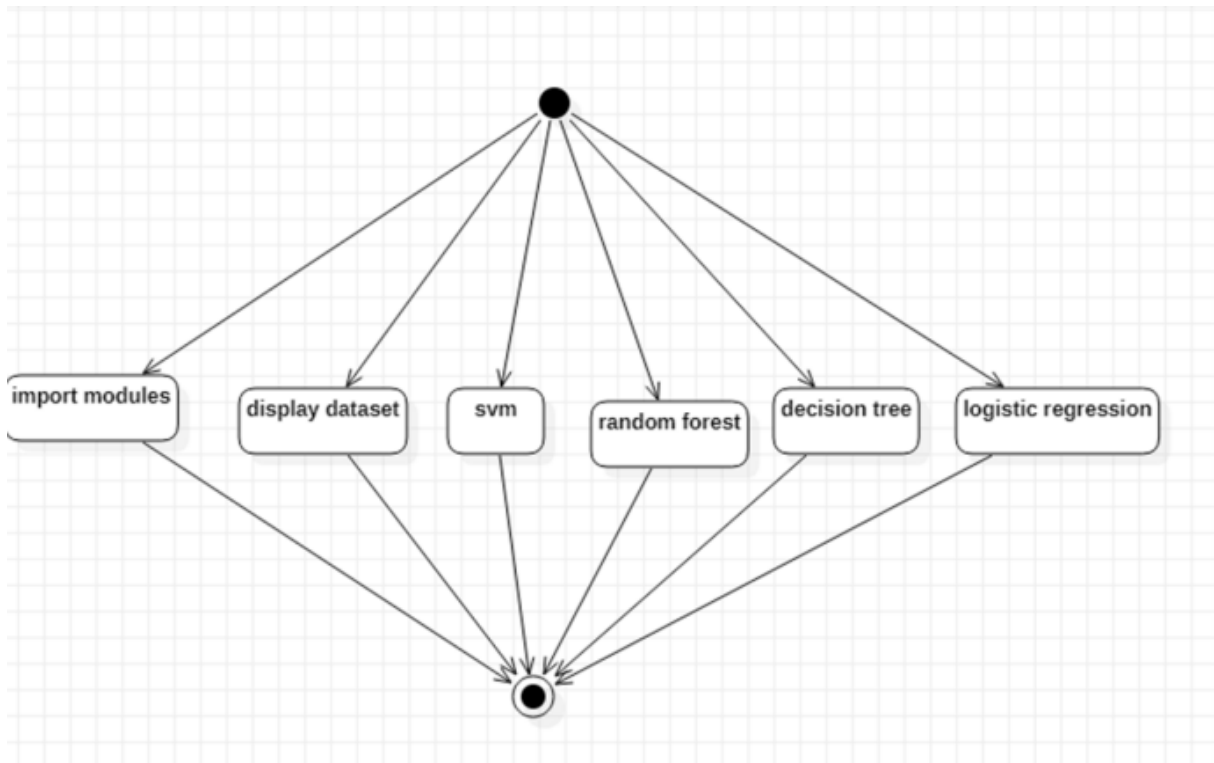


Fig 4.2.4 Activity Diagram

5. IMPLEMENTATION

5.1 DATA COLLECTION

The dataset used in this project was sourced from Kaggle, containing detailed diagnostic information about breast cancer patients. The dataset features various attributes such as cell radius, texture, perimeter, area, and smoothness, which are essential for distinguishing between benign and malignant tumors. The dataset was in a labeled format, with the target variable indicating whether the tumor was benign (non-cancerous) or malignant (cancerous).

The data format was CSV, making it easy to integrate into the Python-based data processing pipelines. A significant amount of time was spent understanding the dataset's features and ensuring its quality before moving on to the preprocessing stage.

5.2 DATA PREPROCESSING

Data preprocessing was a critical step in ensuring the data was clean and suitable for model training. The preprocessing phase included:

- i. **Data Cleaning:** Missing values, if any, were handled appropriately. Outliers and irrelevant features were removed to prevent noise from affecting model performance.
- ii. **Data Transformation:** The features were standardized using techniques such as normalization and scaling. This ensured that all features were on a similar scale, which is important for models like SVM and KNN.
- iii. **Train-Test Split:** The dataset was split into training and testing sets using an 80-20 ratio to ensure that the model could be evaluated effectively on unseen data.
- iv. **Feature Selection:** Only the most relevant features that contributed to the prediction were selected. This helped in reducing dimensionality, improving model performance, and speeding up the training process.

5.3 MODEL TRAINING

Multiple machine learning algorithms were employed to train the model, each chosen for its strengths in classification tasks. The algorithms used include:

- i. **Logistic Regression:** A basic yet powerful classification algorithm suitable for binary outcomes like cancer diagnosis.
- ii. **Decision Tree:** A decision support tool that uses a tree-like model to make decisions based on the input data.
- iii. **Random Forest:** An ensemble method that aggregates the output of multiple decision trees to improve accuracy and reduce overfitting.
- iv. **Support Vector Machine (SVM):** A powerful classification algorithm that works by finding the optimal hyperplane to separate classes.
- v. **K-Nearest Neighbors (KNN):** A simple algorithm that classifies a data point based on how its neighbors are classified.
- vi. **XGBoost:** An optimized gradient boosting algorithm known for its speed and performance in classification tasks.

Each model was trained using the training data, and hyperparameters were fine-tuned using cross-validation to ensure optimal performance. The models were evaluated on the test set to assess how well they generalized to new, unseen data.

5.4 MODEL EVALUATION

The evaluation of the trained models was done using several performance metrics:

- i. **Accuracy:** The proportion of correct predictions out of the total predictions.
- ii. **Precision, Recall, and F1 Score:** These metrics helped in understanding the balance between false positives and false negatives, which is crucial in medical diagnosis where the cost of misclassification can be high.

- iii. **Confusion Matrix:** A confusion matrix was used to visualize the performance of the classification models, giving insights into the number of true positives, true negatives, false positives, and false negatives.
- iv. **Cross-Validation:** K-fold cross-validation was used to mitigate overfitting and to ensure that the model performs well across different subsets of the data.
- v. The best-performing models were Random Forest and XGBoost, which achieved high accuracy and balanced precision and recall. These models were selected for deployment.

5.5 DEPLOYMENT USING FLASK

The deployment of the breast cancer detection model using Flask involved integrating the machine learning model into a web application to make it accessible and usable for end-users. Flask, a lightweight Python framework, was chosen for its simplicity and flexibility in handling web requests and managing API endpoints. The machine learning model, likely serialized using libraries like pickle, was loaded into the Flask application, enabling the web server to handle incoming data, process it using the model, and generate predictions in real-time. This approach facilitated the seamless interaction between the user interface and the backend model, ensuring efficient data flow and response.

The web application interface was designed to be intuitive and user-friendly, allowing users, including those without technical expertise, to input medical information such as features derived from diagnostic tests. The interface, built using HTML, CSS, and JavaScript, communicated with the Flask backend to send these inputs for processing. Upon receiving the data, the Flask server pre-processed it to match the model's expected format, ran the prediction, and then returned the results to the user interface. This real-time feedback mechanism provided immediate diagnostic insights, making it a practical tool for early detection and decision support.

5.6 SAMPLE CODE

5.6.1 BACKEND MODEL

```
# Importing necessary libraries

import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score

# Class for Patient data
class Patient:

    def __init__(self, id, features):
        self.id = id
        self.features = features

# Class for the Prediction Model
class PredictionModel:

    def __init__(self):
        self.model = RandomForestClassifier()

# Method to train the model
    def train_model(self, X_train, y_train):
        self.model.fit(X_train, y_train)

# Method to predict based on patient data
    def predict(self, patient):
        return self.model.predict([patient.features])

# Method to evaluate the model
    def evaluate_model(self, X_test, y_test):
        predictions = self.model.predict(X_test)
```

```

        return accuracy_score(y_test, predictions)

# Load dataset (Breast Cancer dataset from sklearn)
    from sklearn.datasets import load_breast_cancer
    data = load_breast_cancer()

# Preparing data
    X = pd.DataFrame(data.data, columns=data.feature_names)
    y = pd.Series(data.target)

# Splitting data
    X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Creating and training the prediction model
    model = PredictionModel()
    model.train_model(X_train, y_train)

# Sample patient data
    sample_patient = Patient(id=1, features=X_test.iloc[0])

# Predicting for the sample patient
    prediction = model.predict(sample_patient)
    print(f"Prediction for patient {sample_patient.id}: {'Malignant' if prediction == 0 else
    'Benign'}")

# Evaluating the model
    accuracy = model.evaluate_model(X_test, y_test)
    print(f"Model Accuracy: {accuracy * 100:.2f}%")

```


5.6.2 MIDDLEWARE CODE

Saving pickle file

```
import pickle

from sklearn.ensemble import RandomForestClassifier
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split

# Load dataset
data = load_breast_cancer()
    X = data.data
    y = data.target

# Split data
    X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Train a model
model = RandomForestClassifier()
model.fit(X_train, y_train)

# Save the model to a .pkl file
with open('breast_cancer_model.pkl', 'wb') as file:
    pickle.dump(model, file)

print("Model saved to 'breast_cancer_model.pkl'")
```

Loading the Model from a .pkl File

```
import pickle

from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
```

```

# Load dataset
data = load_breast_cancer()
X = data.data
y = data.target

# Split data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Load the model from the .pkl file
with open('breast_cancer_model.pkl', 'rb') as file:
    model = pickle.load(file)

# Evaluate the model
predictions = model.predict(X_test)
accuracy = accuracy_score(y_test, predictions)
print(f"Model Accuracy: {accuracy * 100:.2f}%")

```

Flask Code:

```

Importing essential libraries
from flask import Flask, render_template, request
import joblib
import numpy as np
model = joblib.load('breast_cancer.pkl')
app = Flask(__name__)

@app.route('/')
def home():
    return render_template('newinput.html')

@app.route('/predict', methods=['POST'])

```

```

def predict():
    if request.method == 'POST':
        # Extracting features from the form data
        texture_mean = float(request.form['texture_mean'])
        smoothness_mean = float(request.form['smoothness_mean'])
        compactness_mean = float(request.form['compactness_mean'])
        concave_points_mean = float(request.form['concave_points_mean'])
        symmetry_mean = float(request.form['symmetry_mean'])
        fractal_dimension_mean = float(request.form['fractal_dimension_mean'])
        texture_se = float(request.form['texture_se'])
        area_se = float(request.form['area_se'])
        smoothness_se = float(request.form['smoothness_se'])
        compactness_se = float(request.form['compactness_se'])
        concavity_se = float(request.form['concavity_se'])
        concave_points_se = float(request.form['concave_points_se'])
        symmetry_se = float(request.form['symmetry_se'])
        fractal_dimension_se = float(request.form['fractal_dimension_se'])
        texture_worst = float(request.form['texture_worst'])
        area_worst = float(request.form['area_worst'])
        smoothness_worst = float(request.form['smoothness_worst'])
        compactness_worst = float(request.form['compactness_worst'])
        concavity_worst = float(request.form['concavity_worst'])
        concave_points_worst = float(request.form['concave_points_worst'])
        symmetry_worst = float(request.form['symmetry_worst'])
        fractal_dimension_worst = float(request.form['fractal_dimension_worst'])
        # Create an array of features

```

```

data = np.array([[texture_mean, smoothness_mean, compactness_mean,
                  concave_points_mean, symmetry_mean, fractal_dimension_mean
                  , texture_se, area_se, smoothness_se, compactness_se, concavity_se,
                  concave_points_se, symmetry_se, fractal_dimension_se, texture_worst,
                  area_worst, smoothness_worst, compactness_worst, concavity_worst,
                  concave_points_worst, symmetry_worst, fractal_dimension_worst]])

# Making prediction
my_prediction = model.predict(data)

# Determine suggestion based on prediction
if my_prediction == 1:
    suggestion = "It is recommended to consult with a healthcare professional for
                  further evaluation and possible treatment options."
else:
    suggestion = "No further action is required at this time, but regular screenings
                  are recommended."

# Returning the prediction and suggestion as response
return render_template('newresult.html', prediction=my_prediction,
                       suggestion=suggestion)

if __name__ == '__main__':
    app.run(debug=True)

```

5.6.3 HTML CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Breast Cancer Prediction</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      margin: 0;
      padding: 0;
      background-image:
url('https://www.uft.org/sites/default/files/styles/large/public/photos/BC_hero-no-
text_crop.jpg?itok=VRkL49v-');
      background-size: cover;
      background-position: center;
      background-repeat: no-repeat;
      color: #333;
    }

    <h3>Breast Cancer Awareness</h3>
    <p>Raising awareness about breast cancer can lead to early detection and better
outcomes. Here are some key points to remember:</p>
    <ul>
      <li>Regular self-examinations and mammograms are crucial.</li>
      <li>Maintain a healthy lifestyle to reduce risk factors.</li>
```

```

        <li>Stay informed about the latest research and treatment options.</li>
        <li>Support breast cancer research and awareness campaigns.</li>
    </ul>
</div>

<div class="prevention">
    <h3>Prevention is Better Than Cure</h3>

    <p>Adopting preventive measures can significantly reduce the risk of breast cancer.
    Regular check-ups, a healthy diet, physical activity, and avoiding known risk factors are key
    steps in prevention.</p>
</div>

<h1>Breast Cancer Prediction</h1>
<form action="/predict" method="post">
    <label for="area_worst">Area Worst:</label>
    <input type="number" id="area_worst" name="area_worst" step="any" required>

    <label for="concave_points_mean">Concave Points Mean:</label>
    <input type="number" id="concave_points_mean" name="concave_points_mean"
step="any" required>

    <label for="perimeter_worst">Perimeter Worst:</label>
    <input type="number" id="perimeter_worst" name="perimeter_worst" step="any"
required>

    <label for="concave_points_worst">Concave Points Worst:</label>
    <input type="number" id="concave_points_worst" name="concave_points_worst"
step="any" required>

```

```

<label for="radius_worst">Radius Worst:</label>
<input type="number" id="radius_worst" name="radius_worst" step="any" required>

<label for="perimeter_mean">Perimeter Mean:</label>
<input type="number" id="perimeter_mean" name="perimeter_mean" step="any"
required>

<label for="concavity_mean">Concavity Mean:</label>
<input type="number" id="concavity_mean" name="concavity_mean" step="any"
required>

<label for="concavity_worst">Concavity Worst:</label>
<input type="number" id="concavity_worst" name="concavity_worst" step="any"
required>

<label for="radius_mean">Radius Mean:</label>
<input type="number" id="radius_mean" name="radius_mean" step="any" required>

<label for="area_mean">Area Mean:</label>
<input type="number" id="area_mean" name="area_mean" step="any" required>

<button type="submit">Predict</button>
</form>
</body>
</html>

```

GITHUB LINK: [GitHub - bhavanisai2004/breastcancerprediction](https://github.com/bhavanisai2004/breastcancerprediction)

6. TESTING

6.1 SOFTWARE TESTING

Software testing is crucial in the development of the breast cancer detection system to ensure its accuracy and reliability. The aim is to validate that the system meets the specified requirements and performs effectively. Testing for this project involves several types of testing methodologies:

1. Unit Testing

Unit testing focuses on verifying the functionality of individual components or modules in isolation. For the breast cancer detection system, key unit tests include:

- i. **Feature Extraction:** Testing the feature extraction functions to ensure they correctly process and normalize input data from the user.
- ii. **Model Prediction:** Verifying that the machine learning model correctly processes input features and produces accurate predictions based on the trained model.
- iii. **Data Input Handling:** Ensuring that the system correctly handles and parses user inputs from the web interface.

2. Integration Testing

Integration testing evaluates how well different components of the system work together.

Important integration tests include:

- i. **User Interface and Prediction:** Testing the interaction between the user interface and the prediction engine to confirm that user inputs are correctly sent to the model and results are accurately displayed.
- ii. **Model Integration:** Verifying that the model integration is smooth, ensuring that predictions from the model are properly handled by the application and displayed to the user.

- iii. **Data Flow:** Checking the flow of data from user input through the model prediction to the output, ensuring seamless processing and accurate results.

3. Black Box Testing

Black box testing assesses the system's functionality without delving into its internal code.

Key black box tests for the breast cancer detection system include:

- i. **Input Validation:** Testing the web forms to ensure they correctly handle various input scenarios, including valid and invalid data, and provide appropriate feedback.
- ii. **Prediction Accuracy:** Evaluating the system's ability to make accurate predictions based on known test cases with labeled data.
- iii. **User Interface:** Verifying that the user interface displays results correctly and provides clear instructions and feedback to users.

4. White Box Testing

White box testing involves examining the internal code and logic of the system. Important white box tests include:

- i. **Model Code Verification:** Reviewing the code that handles model predictions to ensure it correctly interfaces with the machine learning model and processes inputs accurately.
- ii. **Error Handling:** Testing error handling routines in the code to ensure that exceptions and edge cases are managed properly.
- iii. **Optimization Checks:** Checking the efficiency and performance of the data processing and prediction code to ensure optimal performance.

6.2 TEST CASES

S. No	Test Case Description	Expected Result	Test Result
1	Test model training with valid dataset	The system should successfully train the model and display the training accuracy and loss.	Success
2	Test prediction with valid input features	The system should predict whether the breast cancer is malignant or benign and return the result with a probability score.	Success
3	Test prediction with missing input features	The system should display an error message indicating that all input features are required.	Success
4	Test accuracy of the prediction model with test data	The system should evaluate the accuracy of the model using the test dataset and return the accuracy score.	Success
5	Test model's ability to detect malignant cases accurately	The system should correctly classify malignant cases with high precision and recall.	Success
6	Test model's ability to detect benign cases accurately	The system should correctly classify benign cases with high precision and recall.	Success
7	Test confusion matrix generation after model evaluation	The system should generate a confusion matrix showing the true positive, true negative, false positive, and false negative counts.	Success
8	Test ROC curve generation after model evaluation	The system should generate a Receiver Operating Characteristic (ROC) curve and calculate the Area Under Curve (AUC) score.	Success

7. OUTPUTS

7.1 EXPLORATORY DATA ANALYSIS

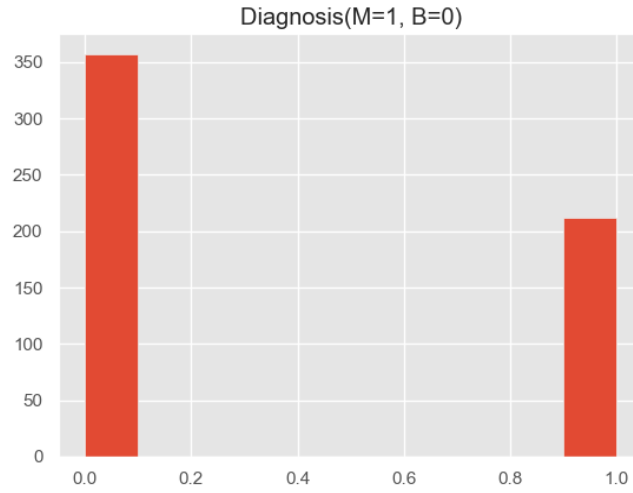


Fig 7.1 Histogram

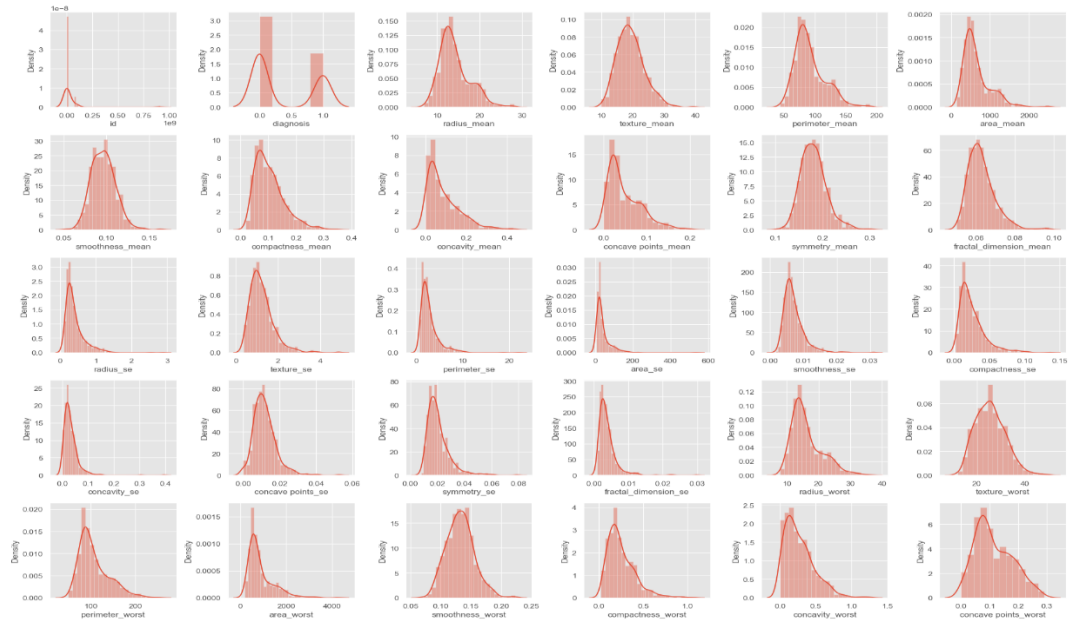


Fig 7.2 Density Plot

The histogram illustrates the distribution of breast cancer diagnoses in the dataset, where 'M' (Malignant) is represented as 1 and 'B' (Benign) as 0. The plot shows a higher frequency of benign cases compared to malignant ones.

The density plot visualizes the distribution of each feature in the dataset across all columns. With 30 plots arranged in a 5x6 grid, the graphs show the probability distribution of each feature, helping to identify any skewness or patterns within the data.

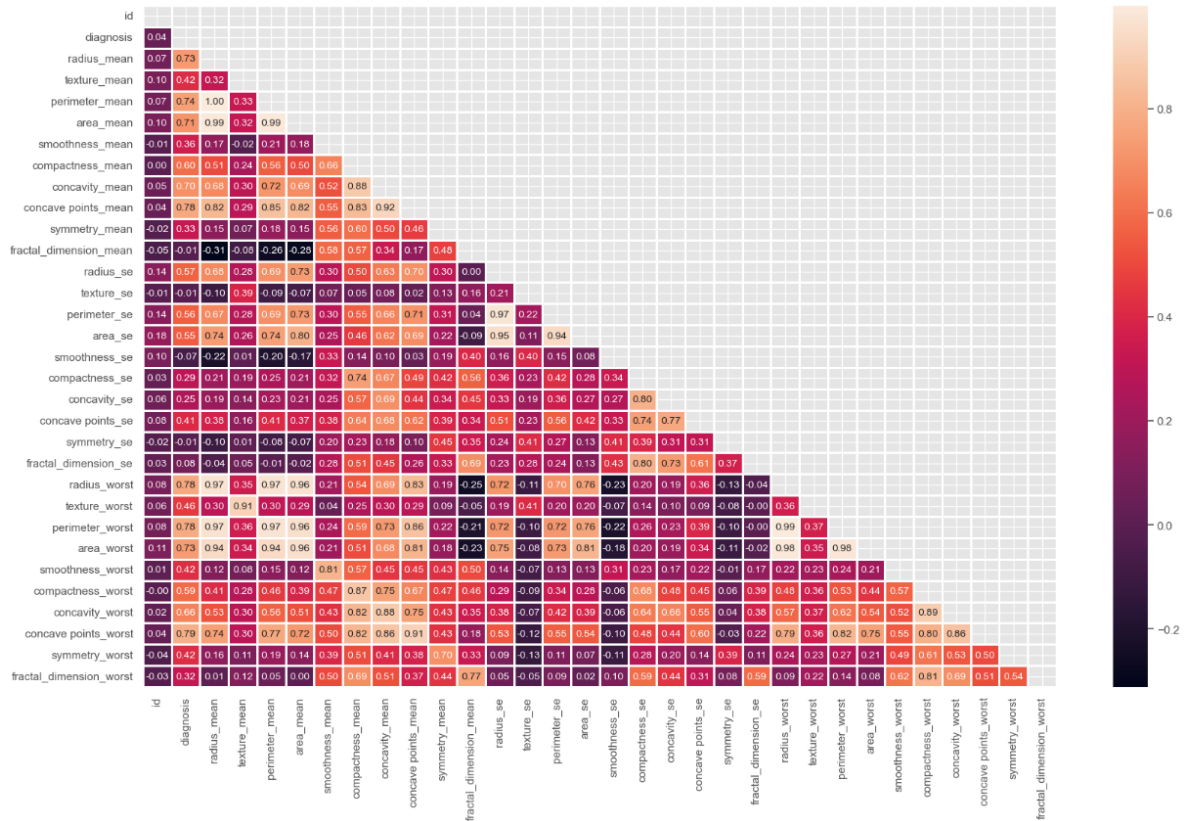


Fig 7.3 HeatMap

The heatmap displays the correlation matrix of the dataset, showing the relationships between different features. The mask hides the upper triangle for clarity, and the annotations provide precise correlation values. Strong positive or negative correlations can be easily identified, helping in feature selection for model building.

	Model	Score
2	SVM	98.25
4	Random Forest Classifier	98.25
0	Logistic Regression	96.49
5	Gradient Boosting Classifier	96.49
1	KNN	95.61
6	XgBoost	95.61
3	Decision Tree Classifier	92.98

Fig 7.4 Comparision frame

DataFrame to compare the performance scores of various machine learning models used for breast cancer detection. The models, such as Logistic Regression, KNN, SVM, Decision Tree, and others, are evaluated based on their accuracy scores, with the DataFrame being sorted in descending order to highlight the best-performing model.

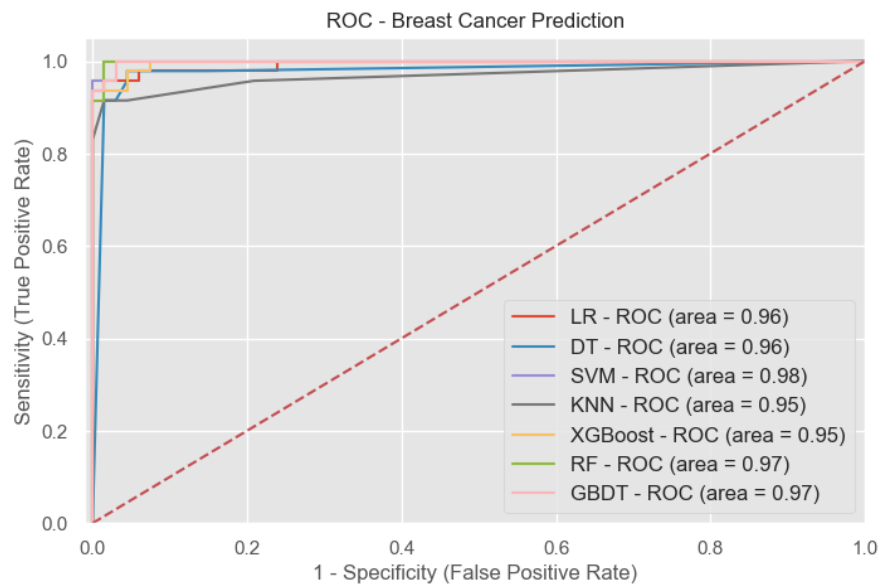


Fig 7.5 ROC curve

ROC (Receiver Operating Characteristic) curves for multiple machine learning models, such as Logistic Regression, Decision Tree, SVM, KNN, XGBoost, Random Forest, and Gradient Boosting, used for breast cancer prediction. The ROC curves illustrate the trade-off between sensitivity (True Positive Rate) and 1-specificity (False Positive Rate). Each curve is plotted along with its AUC (Area Under Curve) score, providing a visual comparison of model performance. The higher the AUC, the better the model distinguishes between classes.

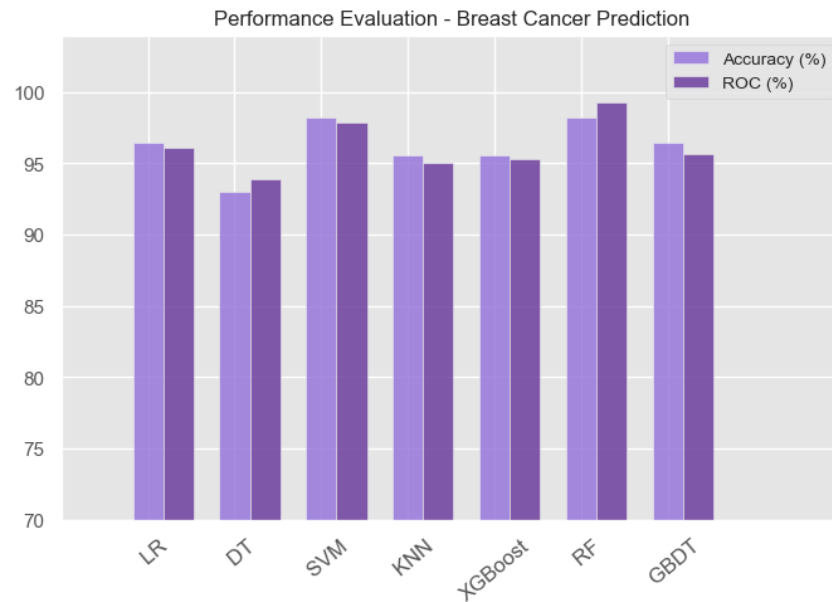


Fig 7.6 Performance evaluation graph

7.2 OUTPUT SCREENS

Breast Cancer Awareness

Raising awareness about breast cancer can lead to early detection and better outcomes. Here are some key points to remember:

- Regular self-examinations and mammograms are crucial.
- Maintain a healthy lifestyle to reduce risk factors.
- Stay informed about the latest research and treatment options.
- Support breast cancer research and awareness campaigns.

Prevention is Better Than Cure

Adopting preventive measures can significantly reduce the risk of breast cancer. Regular check-ups, a healthy diet, physical activity, and avoiding known risk factors are key steps in prevention.

Breast Cancer Prediction

Area Worst:
2019

Concave Points Mean:
0.1471

Fig 7.7 Webpage input

Perimeter Worst:
184.6

Concave Points Worst:
0.2654

Radius Worst:
25.38

Perimeter Mean:
122.8

Concavity Mean:
0.3001

Concavity Worst:
0.7119

Radius Mean:
17.99

Area Mean:
1001

Predict

Fig 7.8 Webpage input

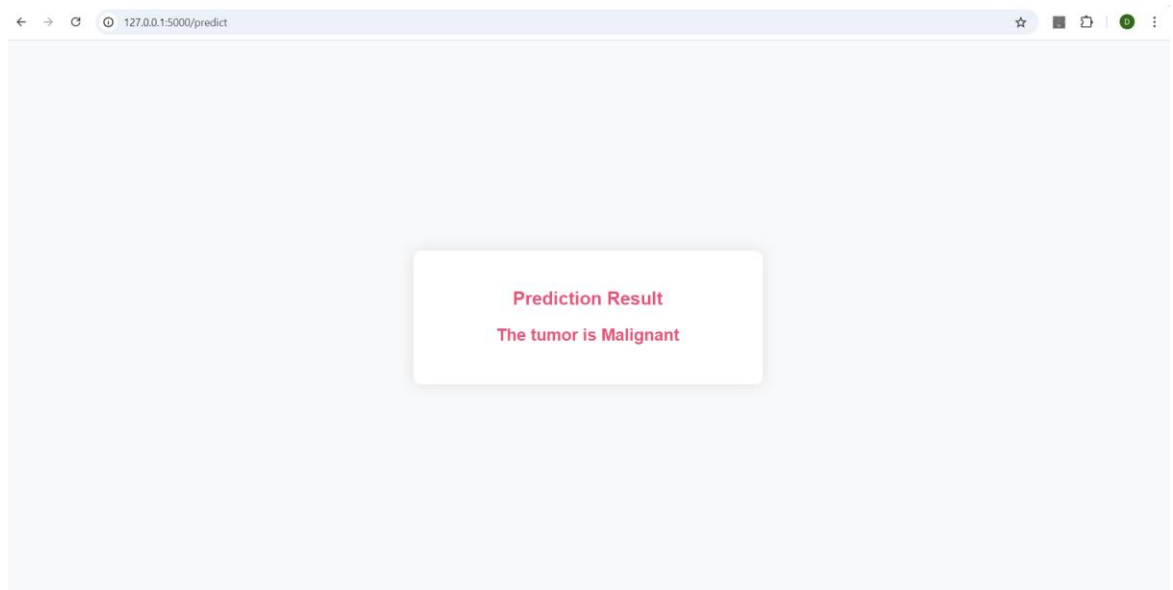


Fig 7.9 Output displaying predictions

For Case-2

A screenshot of a web browser window with the address bar showing "127.0.0.1:5000". The page has a pink background with a breast cancer illustration. The title "Breast Cancer Prediction" is in red. The form contains several input fields with the following labels and values: "Area Worst: 0.9", "Concave Points Mean: 0.1471", "Perimeter Worst: 7.9", "Concave Points Worst: 3.6", "Radius Worst: 6.89", "Perimeter Mean: 5.67", "Concavity Mean: 8.98", "Concavity Worst: 8.90", and "Radius Mean:".

Input Field	Value
Area Worst:	0.9
Concave Points Mean:	0.1471
Perimeter Worst:	7.9
Concave Points Worst:	3.6
Radius Worst:	6.89
Perimeter Mean:	5.67
Concavity Mean:	8.98
Concavity Worst:	8.90
Radius Mean:	

Fig 7.9 Input2

A screenshot of a web application interface for predicting tumor status. The interface is displayed in a browser window with the address bar showing "127.0.0.1:5000". The page has a light blue header and a white main content area. On the left, there is a vertical sidebar with a pink background and a white circular logo. The main content area contains a form with several input fields, each with a label and a value. The labels and values are: "Perimeter Worst: 7.9", "Concave Points Worst: 3.6", "Radius Worst: 6.89", "Perimeter Mean: 5.67", "Concavity Mean: 8.98", "Concavity Worst: 8.90", "Radius Mean: 4.678", and "Area Mean: 5.89". At the bottom of the form is a red button labeled "Predict".

Feature	Value
Perimeter Worst	7.9
Concave Points Worst	3.6
Radius Worst	6.89
Perimeter Mean	5.67
Concavity Mean	8.98
Concavity Worst	8.90
Radius Mean	4.678
Area Mean	5.89

Predict

Fig 7.10 Input2

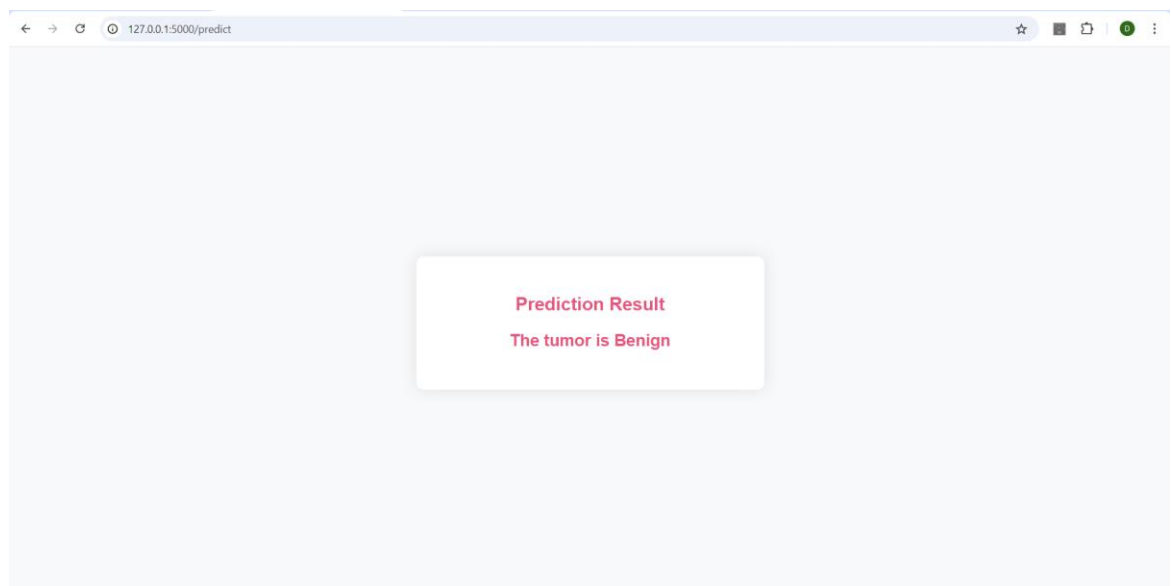


Fig 7.11 Output2

8. CONCLUSION

8.1 CONCLUSION

Breast cancer detection using machine learning represents a significant advancement in healthcare, offering the potential for early diagnosis, which is critical for improving patient outcomes. By leveraging powerful algorithms, the system can analyze medical data to accurately differentiate between malignant and benign cases, enabling timely interventions. Key metrics such as accuracy, precision, recall, and the ROC curve provide confidence in the model's ability to predict breast cancer with a high degree of reliability.

This approach not only supports healthcare professionals by providing an additional tool for diagnosis but also enhances the efficiency of medical processes by reducing human error. As more data becomes available and models are continuously refined, the potential for improving breast cancer detection and treatment outcomes will grow, paving the way for personalized healthcare solutions.

8.2 FURTHER ENHANCEMENTS

- i. **Deep Learning Integration:** Incorporating deep learning techniques like CNNs can significantly improve detection accuracy, especially for image-based data such as mammograms or histopathology slides.
- ii. **Explainable AI:** Adding explainable AI tools will help clinicians understand the decision-making process of the model, increasing trust and facilitating its adoption in clinical settings.
- iii. **Multi-Modal Data Fusion:** Combining various data sources, such as imaging, genetic data, and medical history, will provide a more comprehensive analysis, improving detection and prediction accuracy.
- iv. **Continuous Learning:** Implementing continuous learning with real-time data updates will help the model stay current with medical advancements and patient data, leading to better performance.

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10. APPENDICES

1. Appendix A: System Requirements

The system requires specific software and hardware to ensure smooth development and operation. On the software side, the project is built using Python, preferably within the Anaconda environment (version 3.8 or later), which simplifies package management. Key libraries like Scikit-learn, TensorFlow/Keras, Pandas, NumPy, and Matplotlib are essential for machine learning, data manipulation, and visualization. Development can be done using Jupyter Notebook or VSCode, depending on the user's preference. If the system involves data storage, MySQL or SQLite may be optionally used. For hardware, a computer with an Intel i5 processor and 8GB of RAM is the minimum requirement, though 16GB is recommended for better performance. Storage needs are modest at 100GB. For more complex tasks, an optional NVIDIA GPU with CUDA support can accelerate machine learning processes.

2. Appendix B: Installation Guide

The installation process begins with downloading and installing Anaconda from its official website, following the installation instructions provided. Once installed, users should set up a virtual environment specific to the project and install the required libraries. After setting up the environment, project files can be downloaded and the dataset correctly placed within the project directory. Finally, users can launch the project using Jupyter Notebook or their preferred IDE and proceed with training and testing the machine learning model.

3. Appendix C: Data Schema

The dataset used in the project consists of various features such as `radius_mean` and `texture_mean`, which represent specific measurements from the breast cancer data. The target variable is the diagnosis, which indicates whether the tumor is malignant or benign.

If a database is used, there are two primary tables: the Patients Table, which stores essential details like patient ID, name, and diagnosis, and the Test Results Table, which logs the model's predictions and probabilities for future reference.

4. Appendix D: User Manual

Different roles interact with the system. Data scientists manage data preparation, model training, and evaluation, ensuring that the machine learning pipeline runs smoothly. Clinicians, on the other hand, use the trained model to make predictions and review the results for medical decision-making. Common tasks include loading the dataset to train the model, inputting patient data to make cancer predictions, and evaluating model performance by analyzing accuracy metrics and confusion matrices.

5. Appendix E: Troubleshooting

Some common issues that users may encounter include missing modules, where a required library is not installed, missing data if the dataset is not properly loaded, or model training failures due to incompatible configurations. Troubleshooting these issues involves checking the environment setup, ensuring all necessary libraries and datasets are present, and verifying the model parameters.

6. Appendix F: Glossary

A glossary of key terms is provided for users unfamiliar with technical jargon. Scikit-learn refers to a popular machine learning library for Python, while Pandas is a library used for data manipulation. The ROC Curve is a graphical representation of a classifier's performance, and the Confusion Matrix is a table used to evaluate model prediction accuracy. Jupyter Notebook is an interactive tool used for coding and analyzing data in Python.

11. PLAGIARISM



Fig 11.1 Plagiarism Report